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# Complex Numbers

JEE Main Mathematics · Previous Year Questions with Answer Key

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INSIDE THIS PDF

| | |
|---|---|
| MCQ (Single Correct Answer) | 182 questions |
| Numerical | 38 questions |
| Answer key | at the end |

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## MCQ (Single Correct Answer) 182 questions

**Q1.** JEE Main 2026 (Online) 8th April Evening Shift

The number of values of $z \in \mathbb{C}$, satisfying the equations $|z-(4+8i)| = \sqrt{10}$ and $|z-(3+5i)| + |z-(5+11i)| = 4\sqrt{5}$, is :

(A) 0 (B) 2 (C) 1 (D) 4

Open solution

**Q2.** JEE Main 2026 (Online) 6th April Evening Shift

Let $S = \left\{z \in \mathbb{C} : z^2 + \sqrt{6}iz - 3 = 0\right\}$. Then $\sum_{z \in S} z^8$ is equal to :

(A) 162 (B) 184 (C) 262 (D) 324

Open solution

**Q3.** JEE Main 2026 (Online) 6th April Morning Shift

Let the set of all values of $k \in \mathbb{R}$ such that the equation $z(\bar{z}+2+i) + k(2+3i) = 0, z \in \mathrm{C}$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha+\beta)$ is equal to:

(A) -10 (B) -8 (C) $10\sqrt{13}$ (D) $8\sqrt{13}$

Open solution

**Q4.** JEE Main 2026 (Online) 5th April Evening Shift

Let $z_1, z_2 \in \mathbb{C}$ be the distinct solutions of the equation $z^2 + 4z - (1+12i) = 0$.

Then $|z_1|^2 + |z_2|^2$ is equal to :

(A) 18 (B) 22 (C) 29 (D) 34

Open solution

**Q5.** JEE Main 2026 (Online) 5th April Morning Shift

Let $a, b \in \mathbb{C}$. Let $\alpha, \beta$ be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$, then $\left(\beta^3 - \alpha^3\right)^2$ is equal to:

(A) 160 (B) 176 (C) 194 (D) 187

Open solution

**Q6.** JEE Main 2026 (Online) 4th April Evening Shift

Let $\mathrm{S} = \left\{z \in \mathrm{C} : z^2 + 4z + 16 = 0\right\}$. Then $\sum_{z \in \mathrm{S}} |z + \sqrt{3}\mathrm{i}|^2$ is equal to :

(A) 42 (B) 23 (C) 27 (D) 38

Open solution

**Q7.** JEE Main 2026 (Online) 4th April Morning Shift

Let $z$ be a complex number such that $|z+2| = |z-2|$ and $\arg\left(\frac{z+3}{z-i}\right) = \frac{\pi}{4}$. Then $|z|^2$ is equal to:

(A) 9 (B) 4 (C) 5 (D) 1

Open solution

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**Q8.** JEE Main 2026 (Online) 2nd April Evening Shift

Let the circles $C_1 : |z| = r$ and $C_2 : |z - 3 - 4i| = 5$, $z \in \mathbb{C}$, be such that $C_2$ lies within $C_1$. If $z_1$ moves on $C_1$, $z_2$ moves on $C_2$ and $\min |z_1 - z_2| = 2$, then $\max |z_1 - z_2|$ is equal to :

(A) 12 (B) 17 (C) 22 (D) 24

Open solution

**Q9.** JEE Main 2026 (Online) 2nd April Morning Shift

Let $x$ and $y$ be real numbers such that $50\left(\frac{2x}{1+3i} - \frac{y}{1-2i}\right) = 31 + 17i, i = \sqrt{-1}$. Then the value of $10(x - 3y)$ is :

(A) 20 (B) 31 (C) 35 (D) 75

Open solution

**Q10.** JEE Main 2026 (Online) 28th January Evening Shift

Let

$A = \{z \in \mathbb{C} : |z - 2| \leq 4\}$ and

$B = \{z \in \mathbb{C} : |z - 2| + |z + 2| = 5\}$.

Then the max $\{|z_1 - z_2| : z_1 \in A$ and $z_2 \in B\}$ is :

(A) $\frac{17}{2}$ (B) 8 (C) 9 (D) $\frac{15}{2}$

Open solution

**Q11.** JEE Main 2026 (Online) 28th January Morning Shift

Let $z$ be a complex number such that $|z - 6| = 5$ and $|z + 2 - 6i| = 5$. Then the value of $z^3 + 3z^2 - 15z + 141$ is equal to :

(A) 61 (B) 37 (C) 42 (D) 50

Open solution

**Q12.** JEE Main 2026 (Online) 24th January Morning Shift

Let $S = \left\{z \in \mathbb{C} : \left|\frac{z-6i}{z-2i}\right| = 1 \text{ and } \left|\frac{z-8+2i}{z+2i}\right| = \frac{3}{5}\right\}$.

Then $\sum\limits_{z \in S} |z|^2$ is equal to :

(A) 413 (B) 398 (C) 385 (D) 423

Open solution

**Q13.** JEE Main 2026 (Online) 23rd January Evening Shift

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}, i = \sqrt{-1}$, then $\left(z^{201} - i\right)^8$ is equal to

(A) 1 (B) 0 (C) -1 (D) 256

Open solution

**Q14.** JEE Main 2026 (Online) 23rd January Morning Shift

Let $S = \{z : 3 \leqslant |2z - 3(1 + i)| \leqslant 7\}$ be a set of complex numbers.

Then $\text{Min}_{z \in S} \left|\left(z + \frac{1}{2}(5 + 3i)\right)\right|$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{5}{2}$ (C) 2 (D) $\frac{3}{2}$

Open solution

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**Q15.** JEE Main 2026 (Online) 22nd January Evening Shift

Let $S = \{z \in \mathbb{C} : 4z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} |z|^2$ is equal to:

(A) $\frac{5}{64}$ (B) $\frac{1}{16}$ (C) $\frac{7}{64}$ (D) $\frac{3}{16}$

Open solution

**Q16.** JEE Main 2026 (Online) 21st January Evening Shift

Let $z$ be the complex number satisfying $|z - 5| \leq 3$ and having maximum positive principal argument.

Then $34\left|\frac{5z-12}{5iz+16}\right|^2$ is equal to:

(A) 20 (B) 26 (C) 12 (D) 16

Open solution

**Q17.** JEE Main 2026 (Online) 21st January Morning Shift

If $x^2 + x + 1 = 0$, then the value of $\left(x + \frac{1}{x}\right)^4 + \left(x^2 + \frac{1}{x^2}\right)^4 + \left(x^3 + \frac{1}{x^3}\right)^4 + \ldots + \left(x^{25} + \frac{1}{x^{25}}\right)^4$ is:

(A) 162 (B) 145 (C) 128 (D) 175

Open solution

**Q18.** JEE Main 2025 (Online) 8th April Evening Shift

Let $A = \left\{\theta \in [0, 2\pi] : 1 + 10\,\mathrm{Re}\left(\frac{2\cos\theta + i\sin\theta}{\cos\theta - 3i\sin\theta}\right) = 0\right\}$. Then $\sum_{\theta \in A} \theta^2$ is equal to

(A) $\frac{21}{4}\pi^2$ (B) $6\pi^2$ (C) $\frac{27}{4}\pi^2$ (D) $8\pi^2$

Open solution

**Q19.** JEE Main 2025 (Online) 7th April Evening Shift

If the locus of z $\in \mathbb{C}$, such that $\mathrm{Re}\left(\frac{z-1}{2z+i}\right) + \mathrm{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$, is a circle of radius r and center $(a, b)$, then $\frac{15ab}{r^2}$ is equal to :

(A) 16 (B) 24 (C) 12 (D) 18

Open solution

**Q20.** JEE Main 2025 (Online) 7th April Morning Shift

Among the statements

(S1) : The set $\{z \in \mathbb{C} - \{-i\} : |z| = 1$ and $\frac{z-i}{z+i}$ is purely real $\}$ contains exactly two elements, and

(S2) : The set $\{z \in \mathbb{C} - \{-1\} : |z| = 1$ and $\frac{z-1}{z+1}$ is purely imaginary $\}$ contains infinitely many elements.

(A) both are incorrect (B) both are correct

(C) only (S2) is correct (D) only (S1) is correct

Open solution

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**Q21.** JEE Main 2025 (Online) 4th April Evening Shift

Let the product of $\omega_1 = (8+i)\sin\theta + (7+4i)\cos\theta$ and $\omega_2 = (1+8i)\sin\theta + (4+7i)\cos\theta$ be $\alpha + i\beta$, $i = \sqrt{-1}$. Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively. Then $p + q$ is equal to :

(A) 130 (B) 150 (C) 160 (D) 140

Open solution

**Q22.** JEE Main 2025 (Online) 3rd April Evening Shift

If $z_1, z_2, z_3 \in \mathbb{C}$ are the vertices of an equilateral triangle whose centroid is $z_0$, then $\sum_{k=1}^{3} (z_k - z_0)^2$ is equal to

(A) 0 (B) 1 (C) i (D) -i

Open solution

**Q23.** JEE Main 2025 (Online) 3rd April Morning Shift

Let $z \in C$ be such that $\frac{z^2+3i}{z-2+i} = 2 + 3i$. Then the sum of all possible values of $z^2$ is :

(A) $-19 + 2i$ (B) $-19 - 2i$ (C) $19 - 2i$ (D) $19 + 2i$

Open solution

**Q24.** JEE Main 2025 (Online) 2nd April Morning Shift

Let $z$ be a complex number such that $|z| = 1$. If $\frac{2+k^2 z}{k+\bar{z}} = kz, k \in \mathbf{R}$, then the maximum distance of $k + ik^2$ from the circle $|z - (1+2i)| = 1$ is :

(A) $\sqrt{5} + 1$ (B) 3 (C) $\sqrt{3} + 1$ (D) 2

Open solution

**Q25.** JEE Main 2025 (Online) 29th January Morning Shift

Let $|z_1 - 8 - 2i| \leq 1$ and $|z_2 - 2 + 6i| \leq 2$, $z_1, z_2 \in \mathbb{C}$. Then the minimum value of $|z_1 - z_2|$ is :

(A) 3 (B) 10 (C) 7 (D) 13

Open solution

**Q26.** JEE Main 2025 (Online) 28th January Evening Shift

If $\alpha + i\beta$ and $\gamma + i\delta$ are the roots of $x^2 - (3-2i)x - (2i-2) = 0$, $i = \sqrt{-1}$, then $\alpha\gamma + \beta\delta$ is equal to:

(A) 2 (B) -6 (C) 6 (D) -2

Open solution

**Q27.** JEE Main 2025 (Online) 28th January Morning Shift

Let $O$ be the origin, the point $A$ be $z_1 = \sqrt{3} + 2\sqrt{2}i$, the point $B(z_2)$ be such that $\sqrt{3}|z_2| = |z_1|$ and $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$. Then

(A) area of triangle ABO is $\frac{11}{4}$

(B) area of triangle ABO is $\frac{11}{\sqrt{3}}$

(C) ABO is a scalene triangle

(D) ABO is an obtuse angled isosceles triangle

Open solution

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**Q28.** JEE Main 2025 (Online) 24th January Morning Shift

If $\alpha$ and $\beta$ are the roots of the equation $2z^2 - 3z - 2i = 0$, where $i = \sqrt{-1}$, then $16 \cdot \mathrm{Re}\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right) \cdot \mathrm{Im}\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right)$ is equal to

(A) 441 (B) 312 (C) 409 (D) 398

Open solution

**Q29.** JEE Main 2025 (Online) 23rd January Evening Shift

The number of complex numbers $z$, satisfying $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$, is :

(A) 8 (B) 10 (C) 4 (D) 6

Open solution

**Q30.** JEE Main 2025 (Online) 23rd January Morning Shift

Let $\left|\frac{\bar{z}-i}{2\bar{z}+i}\right| = \frac{1}{3}, z \in C$, be the equation of a circle with center at $C$. If the area of the triangle, whose vertices are at the points $(0, 0), C$ and $(\alpha, 0)$ is 11 square units, then $\alpha^2$ equals:

(A) $\frac{121}{25}$ (B) 100 (C) $\frac{81}{25}$ (D) 50

Open solution

**Q31.** JEE Main 2025 (Online) 22nd January Evening Shift

Let the curve $z(1+i) + \bar{z}(1-i) = 4, z \in C$, divide the region $|z-3| \leq 1$ into two parts of areas $\alpha$ and $\beta$. Then $|\alpha - \beta|$ equals :

(A) $1 + \frac{\pi}{3}$ (B) $1 + \frac{\pi}{6}$ (C) $1 + \frac{\pi}{2}$ (D) $1 + \frac{\pi}{4}$

Open solution

**Q32.** JEE Main 2025 (Online) 22nd January Morning Shift

Let $z_1, z_2$ and $z_3$ be three complex numbers on the circle $|z| = 1$ with $\arg(z_1) = \frac{-\pi}{4}, \arg(z_2) = 0$ and $\arg(z_3) = \frac{\pi}{4}$. If $|z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1|^2 = \alpha + \beta\sqrt{2}, \alpha, \beta \in Z$, then the value of $\alpha^2 + \beta^2$ is :

(A) 41 (B) 29 (C) 24 (D) 31

Open solution

**Q33.** JEE Main 2024 (Online) 9th April Evening Shift

Let $z$ be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z - (6+8i)|$ is equal to

(A) 8 (B) 12 (C) 10 (D) $\infty$

Open solution

**Q34.** JEE Main 2024 (Online) 8th April Evening Shift

The sum of all possible values of $\theta \in [-\pi, 2\pi]$, for which $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is purely imaginary, is equal to :

(A) $4\pi$ (B) $3\pi$ (C) $2\pi$ (D) $5\pi$

Open solution

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**Q35.** JEE Main 2024 (Online) 8th April Morning Shift

Let $z$ be a complex number such that $|z+2|=1$ and $\operatorname{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$. Then the value of $|\operatorname{Re}(\overline{z+2})|$ is

(A) $\frac{2\sqrt{6}}{5}$ (B) $\frac{24}{5}$ (C) $\frac{\sqrt{6}}{5}$ (D) $\frac{1+\sqrt{6}}{5}$

Open solution

**Q36.** JEE Main 2024 (Online) 8th April Morning Shift

If the set $R=\{(a,b): a+5b=42, a,b\in\mathbb{N}\}$ has $m$ elements and $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is

(A) 12 (B) 4 (C) 8 (D) 5

Open solution

**Q37.** JEE Main 2024 (Online) 6th April Evening Shift

If $z_1, z_2$ are two distinct complex numbers such that $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$, then

(A) either $z_1$ lies on a circle of radius $\frac{1}{2}$ or $z_2$ lies on a circle of radius 1.

(B) $z_1$ lies on a circle of radius $\frac{1}{2}$ and $z_2$ lies on a circle of radius 1.

(C) either $z_1$ lies on a circle of radius 1 or $z_2$ lies on a circle of radius $\frac{1}{2}$.

(D) both $z_1$ and $z_2$ lie on the same circle.

Open solution

**Q38.** JEE Main 2024 (Online) 5th April Evening Shift

Let $S_1=\{z\in\mathbf{C}:|z|\le 5\}, S_2=\left\{z\in\mathbf{C}:\operatorname{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right)\ge 0\right\}$ and $S_3=\{z\in\mathbf{C}:\operatorname{Re}(z)\ge 0\}$. Then the area of the region $S_1\cap S_2\cap S_3$ is :

(A) $\frac{125\pi}{24}$ (B) $\frac{125\pi}{6}$ (C) $\frac{125\pi}{12}$ (D) $\frac{125\pi}{4}$

Open solution

**Q39.** JEE Main 2024 (Online) 5th April Morning Shift

Consider the following two statements :

Statement I: For any two non-zero complex numbers $z_1, z_2$, $(|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right|\le 2(|z_1|+|z_2|)$, and

Statement II : If $x, y, z$ are three distinct complex numbers and $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are three positive real numbers such that $\frac{\mathrm{a}}{|y-z|}=\frac{\mathrm{b}}{|z-x|}=\frac{\mathrm{c}}{|x-y|}$, then $\frac{\mathrm{a}^2}{y-z}+\frac{\mathrm{b}^2}{z-x}+\frac{\mathrm{c}^2}{x-y}=1$.

Between the above two statements,

(A) both Statement I and Statement II are incorrect.

(B) Statement I is correct but Statement II is incorrect.

(C) Statement I is incorrect but Statement II is correct.

(D) both Statement I and Statement II are correct.

Open solution

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**Q40.** JEE Main 2024 (Online) 4th April Evening Shift

The area (in sq. units) of the region $S = \{z \in \mathbb{C} : |z-1| \leq 2; (z+\bar{z}) + i(z-\bar{z}) \leq 2, \text{Im}(z) \geq 0\}$ is

(A) $\frac{7\pi}{4}$ (B) $\frac{3\pi}{2}$ (C) $\frac{7\pi}{3}$ (D) $\frac{17\pi}{8}$

Open solution

**Q41.** JEE Main 2024 (Online) 4th April Morning Shift

Let $\alpha$ and $\beta$ be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0, z \in C$. Then $4(\alpha^2 + \beta^2)$ is equal to :

(A) 4 (B) 2 (C) 6 (D) 8

Open solution

**Q42.** JEE Main 2024 (Online) 1st February Evening Shift

If $z$ is a complex number such that $|z| \leqslant 1$, then the minimum value of $\left|z + \frac{1}{2}(3+4i)\right|$ is :

(A) 2 (B) $\frac{5}{2}$ (C) $\frac{3}{2}$ (D) 3

Open solution

**Q43.** JEE Main 2024 (Online) 1st February Morning Shift

Let $S = \{z \in C : |z-1| = 1 \text{ and } (\sqrt{2}-1)(z+\bar{z}) - i(z-\bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that $|z_1| = \max_{z \in S} |z|$ and $|z_2| = \min_{z \in S} |z|$. Then $\left|\sqrt{2}z_1 - z_2\right|^2$ equals :

(A) 1 (B) 4 (C) 3 (D) 2

Open solution

**Q44.** JEE Main 2024 (Online) 31st January Evening Shift

Let $z_1$ and $z_2$ be two complex numbers such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$ Then, $\left|z_1^4 + z_2^4\right|$ equals -

(A) $15\sqrt{15}$ (B) $30\sqrt{3}$ (C) $25\sqrt{3}$ (D) 75

Open solution

**Q45.** JEE Main 2024 (Online) 30th January Evening Shift

If $z$ is a complex number, then the number of common roots of the equations $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to

(A) 0 (B) 2 (C) 1 (D) 3

Open solution

**Q46.** JEE Main 2024 (Online) 30th January Morning Shift

If $z = x + iy, xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $\left|z^2\right|$ is equal to :

(A) 9 (B) $\frac{1}{4}$ (C) 4 (D) 1

Open solution

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**Q47.** JEE Main 2024 (Online) 29th January Evening Shift

Let r and $\theta$ respectively be the modulus and amplitude of the complex number $z = 2 - i\left(2\tan\frac{5\pi}{8}\right)$, then $(r, \theta)$ is equal to

(A) $\left(2\sec\frac{11\pi}{8}, \frac{11\pi}{8}\right)$ (B) $\left(2\sec\frac{3\pi}{8}, \frac{3\pi}{8}\right)$

(C) $\left(2\sec\frac{5\pi}{8}, \frac{3\pi}{8}\right)$ (D) $\left(2\sec\frac{3\pi}{8}, \frac{5\pi}{8}\right)$

Open solution

**Q48.** JEE Main 2024 (Online) 29th January Morning Shift

If $z = \frac{1}{2} - 2i$ is such that $|z+1| = \alpha z + \beta(1+i), i = \sqrt{-1}$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is equal to

(A) 2 (B) $-4$ (C) 3 (D) $-1$

Open solution

**Q49.** JEE Main 2024 (Online) 27th January Morning Shift

If $S = \{z \in C : |z-i| = |z+i| = |z-1|\}$, then, $n(S)$ is :

(A) 1 (B) 2 (C) 3 (D) 0

Open solution

**Q50.** JEE Main 2023 (Online) 15th April Morning Shift

If the set $\left\{\mathrm{Re}\left(\frac{z-\bar{z}+z\bar{z}}{2-3z+5\bar{z}}\right) : z \in \mathbb{C}, \mathrm{Re}(z) = 3\right\}$ is equal to the interval $(\alpha, \beta]$, then $24(\beta - \alpha)$ is equal to :

(A) 36 (B) 27 (C) 42 (D) 30

Open solution

**Q51.** JEE Main 2023 (Online) 13th April Evening Shift

Let $S = \left\{z \in \mathbb{C} : \bar{z} = i\left(z^2 + \mathrm{Re}(\bar{z})\right)\right\}$. Then $\sum_{z \in S} |z|^2$ is equal to :

(A) $\frac{7}{2}$ (B) 4 (C) 3 (D) $\frac{5}{2}$

Open solution

**Q52.** JEE Main 2023 (Online) 12th April Morning Shift

Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1+3i)$ and radius $r = 1$. Let $z_1 = 1 + i$ and the complex number $z_2$ be outside the circle $C$ such that $|z_1 - z_0|\,|z_2 - z_0| = 1$. If $z_0, z_1$ and $z_2$ are collinear, then the smaller value of $|z_2|^2$ is equal to :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{13}{2}$ (D) $\frac{7}{2}$

Open solution

**Q53.** JEE Main 2023 (Online) 11th April Evening Shift

For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : \mathrm{Re}(a + \bar{z}) > \mathrm{Im}(\bar{a} + z)\}$ and $B = \{z \in \mathbb{C} : \mathrm{Re}(a + \bar{z}) < \mathrm{Im}(\bar{a} + z)\}$. Then among the two statements :

(S1): If $\mathrm{Re}(a), \mathrm{Im}(a) > 0$, then the set A contains all the real numbers

(S2) : If $\mathrm{Re}(a), \mathrm{Im}(a) < 0$, then the set B contains all the real numbers,

(A) both are false (B) only (S1) is true

(C) only (S2) is true (D) both are true

Open solution

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**Q54.** JEE Main 2023 (Online) 11th April Morning Shift

Let $w_1$ be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and $w_2$ be the point obtained by the rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then the principal argument of $w_1 - w_2$ is equal to :

(A) $-\pi + \tan^{-1}\frac{8}{9}$ (B) $-\pi + \tan^{-1}\frac{33}{5}$ (C) $\pi - \tan^{-1}\frac{8}{9}$ (D) $\pi - \tan^{-1}\frac{33}{5}$

Open solution

**Q55.** JEE Main 2023 (Online) 10th April Evening Shift

Let $S = \left\{z = x + iy : \frac{2z-3i}{4z+2i} \text{ is a real number}\right\}$. Then which of the following is NOT correct?

(A) $y + x^2 + y^2 \neq -\frac{1}{4}$

(B) $(x, y) = \left(0, -\frac{1}{2}\right)$

(C) $x = 0$

(D) $y \in \left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, \infty\right)$

Open solution

**Q56.** JEE Main 2023 (Online) 10th April Morning Shift

Let the complex number $z = x + iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to :

(A) $\frac{4}{3}$ (B) $\frac{3}{2}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$

Open solution

**Q57.** JEE Main 2023 (Online) 8th April Evening Shift

Let $A = \left\{\theta \in (0, 2\pi) : \frac{1+2i\sin\theta}{1-i\sin\theta} \text{ is purely imaginary }\right\}$. Then the sum of the elements in A is :

(A) $3\pi$ (B) $\pi$ (C) $2\pi$ (D) $4\pi$

Open solution

**Q58.** JEE Main 2023 (Online) 8th April Morning Shift

If for $z = \alpha + i\beta$, $|z + 2| = z + 4(1 + i)$, then $\alpha + \beta$ and $\alpha\beta$ are the roots of the equation :

(A) $x^2 + 2x - 3 = 0$ (B) $x^2 + 3x - 4 = 0$ (C) $x^2 + x - 12 = 0$ (D) $x^2 + 7x + 12 = 0$

Open solution

**Q59.** JEE Main 2023 (Online) 6th April Evening Shift

Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \left\{z \in \mathbb{C} : \text{Re}\left(az^2 + bz\right) = a \text{ and } \text{Re}\left(bz^2 + az\right) = b\right\}$ is equal to :

(A) 0 (B) 2 (C) 1 (D) Infinite

Open solution

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**Q60.** JEE Main 2023 (Online) 1st February Evening Shift

Let $a, b$ be two real numbers such that $ab < 0$. If the complex number $\frac{1+ai}{b+i}$ is of unit modulus and $a + ib$ lies on the circle $|z - 1| = |2z|$, then a possible value of $\frac{1+[a]}{4b}$, where $[t]$ is greatest integer function, is :

Open solution

(A) $\left(\frac{1+\sqrt{7}}{4}\right)$ (B) $\frac{1}{2}$

(C) 0 (D) $-1$

**Q61.** JEE Main 2023 (Online) 1st February Morning Shift

If the center and radius of the circle $\left|\frac{z-2}{z-3}\right| = 2$ are respectively $(\alpha, \beta)$ and $\gamma$, then $3(\alpha + \beta + \gamma)$ is equal to :

Open solution

(A) 12 (B) 10 (C) 11 (D) 9

**Q62.** JEE Main 2023 (Online) 31st January Evening Shift

The complex number $z = \frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$ is equal to :

Open solution

(A) $\cos\frac{\pi}{12} - i\sin\frac{\pi}{12}$ (B) $\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

(C) $\sqrt{2}i\left(\cos\frac{5\pi}{12} - i\sin\frac{5\pi}{12}\right)$ (D) $\sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$

**Q63.** JEE Main 2023 (Online) 31st January Morning Shift

For all $z \in C$ on the curve $C_1 : |z| = 4$, let the locus of the point $z + \frac{1}{z}$ be the curve $C_2$. Then :

Open solution

(A) the curves $C_1$ and $C_2$ intersect at 4 points

(B) the curve $C_2$ lies inside $C_1$

(C) the curve $C_1$ lies inside $C_2$

(D) the curves $C_1$ and $C_2$ intersect at 2 points

**Q64.** JEE Main 2023 (Online) 29th January Morning Shift

For two non-zero complex numbers $z_1$ and $z_2$, if $\text{Re}(z_1 z_2) = 0$ and $\text{Re}(z_1 + z_2) = 0$, then which of the following are possible?

Open solution

A. $\text{Im}(z_1) > 0$ and $\text{Im}(z_2) > 0$

B. $\text{Im}(z_1) < 0$ and $\text{Im}(z_2) > 0$

C. $\text{Im}(z_1) > 0$ and $\text{Im}(z_2) < 0$

D. $\text{Im}(z_1) < 0$ and $\text{Im}(z_2) < 0$

Choose the correct answer from the options given below :

(A) A and C (B) A and B (C) B and D (D) B and C

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**Q65.** JEE Main 2023 (Online) 25th January Evening Shift

Let $z$ be a complex number such that $\left|\frac{z-2i}{z+i}\right| = 2, z \neq -i$. Then $z$ lies on the circle of radius 2 and centre :

(A) $(0, -2)$ (B) $(0, 0)$ (C) $(0, 2)$ (D) $(2, 0)$

Open solution

**Q66.** JEE Main 2023 (Online) 25th January Morning Shift

Let $z_1 = 2 + 3i$ and $z_2 = 3 + 4i$. The set $S = \left\{z \in \mathbb{C} : |z - z_1|^2 - |z - z_2|^2 = |z_1 - z_2|^2\right\}$ represents a

(A) hyperbola with the length of the transverse axis 7

(B) hyperbola with eccentricity 2

(C) straight line with the sum of its intercepts on the coordinate axes equals $-18$

(D) straight line with the sum of its intercepts on the coordinate axes equals $14$

Open solution

**Q67.** JEE Main 2023 (Online) 24th January Evening Shift

The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is

(A) $-\frac{1}{2}\left(1 - i\sqrt{3}\right)$ (B) $-\frac{1}{2}\left(\sqrt{3} - i\right)$

(C) $\frac{1}{2}\left(1 - i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(\sqrt{3} + i\right)$

Open solution

**Q68.** JEE Main 2023 (Online) 24th January Morning Shift

Let $p, q \in \mathbb{R}$ and $\left(1 - \sqrt{3}i\right)^{200} = 2^{199}(p + iq), i = \sqrt{-1}$ then $p + q + q^2$ and $p - q + q^2$ are roots of the equation.

(A) $x^2 + 4x - 1 = 0$ (B) $x^2 - 4x + 1 = 0$ (C) $x^2 + 4x + 1 = 0$ (D) $x^2 - 4x - 1 = 0$

Open solution

**Q69.** JEE Main 2022 (Online) 30th July Evening Shift

Let $z_1 = 1 + 2i, z_2 = 2 + i, \frac{1}{z_1} + \frac{1}{z_2} = \frac{6}{w}$, and $z = \frac{iw}{2 - \overline{w}}$.

Consider the statements

$(S_1)$ $|z| = \frac{5}{\sqrt{17}}$

$(S_2)$ $\arg(z) + \arg(w) = \tan^{-1}\left(\frac{5}{3}\right)$.

Then

(A) both $(S_1)$ and $(S_2)$ are wrong (B) only $(S_1)$ is correct

(C) only $(S_2)$ is correct (D) both $(S_1)$ and $(S_2)$ are correct

Open solution

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**Q70.** JEE Main 2022 (Online) 29th July Evening Shift

If $z \neq 0$ be a complex number such that $\left|z - \frac{1}{z}\right| = 2$, then the maximum value of $|z|$ is :

(A) $\sqrt{2}$ (B) 1 (C) $\sqrt{2} - 1$ (D) $\sqrt{2} + 1$

Open solution

**Q71.** JEE Main 2022 (Online) 29th July Evening Shift

Let $S = \{z = x + iy : |z - 1 + i| \geq |z|, |z| < 2, |z + i| = |z - 1|\}$. Then the set of all values of $x$, for which $w = 2x + iy \in S$ for some $y \in \mathbb{R}$, is :

(A) $\left(-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$ (B) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$

(C) $\left(-\sqrt{2}, \frac{1}{2}\right]$ (D) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

Open solution

**Q72.** JEE Main 2022 (Online) 29th July Morning Shift

If $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to :

(A) 244 (B) 224 (C) 245 (D) 265

Open solution

**Q73.** JEE Main 2022 (Online) 28th July Morning Shift

Let $S_1 = \left\{z_1 \in \mathbf{C} : |z_1 - 3| = \frac{1}{2}\right\}$ and $S_2 = \{z_2 \in \mathbf{C} : |z_2 - |z_2 + 1|| = |z_2 + |z_2 - 1||\}$. Then, for $z_1 \in S_1$ and $z_2 \in S_2$, the least value of $|z_2 - z_1|$ is :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Open solution

**Q74.** JEE Main 2022 (Online) 27th July Evening Shift

Let S be the set of all $(\alpha, \beta), \pi < \alpha, \beta < 2\pi$, for which the complex number $\frac{1 - i\sin\alpha}{1 + 2i\sin\alpha}$ is purely imaginary and $\frac{1 + i\cos\beta}{1 - 2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta} = \sin 2\alpha + i\cos 2\beta, (\alpha, \beta) \in S$. Then $\sum\limits_{(\alpha,\beta)\in S} \left(iZ_{\alpha\beta} + \frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ is equal to :

(A) 3 (B) 3 i (C) 1 (D) 2 − i

Open solution

**Q75.** JEE Main 2022 (Online) 27th July Morning Shift

Let the minimum value $v_0$ of $v = |z|^2 + |z - 3|^2 + |z - 6i|^2, z \in \mathbb{C}$ is attained at $z = z_0$. Then $\left|2z_0^2 - \bar{z}_0^3 + 3\right|^2 + v_0^2$ is equal to :

(A) 1000 (B) 1024 (C) 1105 (D) 1196

Open solution

**Q76.** JEE Main 2022 (Online) 26th July Evening Shift

If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z - i| - |z + 5i| = 0$, then :

(A) $x + 2y - 4 = 0$ (B) $x^2 + y - 4 = 0$ (C) $x + 2y + 4 = 0$ (D) $x^2 - y + 3 = 0$

Open solution

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**Q77.** JEE Main 2022 (Online) 26th July Morning Shift

Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point $z_2$, $\mathrm{Re}(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true?

(A) $\arg z_2 = \pi - \tan^{-1}3$

(B) $\arg(z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$

(C) $|z_2| = \sqrt{10}$

(D) $|2z_1 - z_2| = 5$

Open solution

**Q78.** JEE Main 2022 (Online) 25th July Evening Shift

For $z \in \mathbb{C}$ if the minimum value of $(|z - 3\sqrt{2}| + |z - p\sqrt{2}i|)$ is $5\sqrt{2}$, then a value of $p$ is _____.

(A) 3

(B) $\frac{7}{2}$

(C) 4

(D) $\frac{9}{2}$

Open solution

**Q79.** JEE Main 2022 (Online) 25th July Morning Shift

For $\mathrm{n} \in \mathbf{N}$, let $\mathrm{S_n} = \left\{z \in \mathbf{C} : |z - 3 + 2i| = \frac{\mathrm{n}}{4}\right\}$ and $\mathrm{T_n} = \left\{z \in \mathbf{C} : |z - 2 + 3i| = \frac{1}{\mathrm{n}}\right\}$. Then the number of elements in the set $\{n \in \mathbf{N} : S_n \cap T_n = \phi\}$ is :

(A) 0

(B) 2

(C) 3

(D) 4

Open solution

**Q80.** JEE Main 2022 (Online) 30th June Morning Shift

The real part of the complex number $\frac{(1+2i)^8 \cdot (1-2i)^2}{(3+2i) \cdot \overline{(4-6i)}}$ is equal to :

(A) $\frac{500}{13}$

(B) $\frac{110}{13}$

(C) $\frac{55}{6}$

(D) $\frac{550}{13}$

Open solution

**Q81.** JEE Main 2022 (Online) 29th June Evening Shift

Let arg(z) represent the principal argument of the complex number z. Then, |z| = 3 and arg(z − 1) − arg(z + 1) = $\frac{\pi}{4}$ intersect :

(A) exactly at one point.

(B) exactly at two points.

(C) nowhere.

(D) at infinitely many points.

Open solution

**Q82.** JEE Main 2022 (Online) 29th June Morning Shift

Let $\alpha$ and $\beta$ be the roots of the equation $x^2 + (2i - 1) = 0$. Then, the value of $|\alpha^8 + \beta^8|$ is equal to :

(A) 50

(B) 250

(C) 1250

(D) 1500

Open solution

**Q83.** JEE Main 2022 (Online) 27th June Evening Shift

The number of points of intersection of

$|z - (4 + 3i)| = 2$ and $|z| + |z - 4| = 6$, z ∈ C, is :

(A) 0

(B) 1

(C) 2

(D) 3

Open solution

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**Q84.** JEE Main 2022 (Online) 27th June Morning Shift

The area of the polygon, whose vertices are the non-real roots of the equation $\overline{z} = iz^2$ is :

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Open solution

**Q85.** JEE Main 2022 (Online) 26th June Morning Shift

Let $A = \left\{z \in C : \left|\frac{z+1}{z-1}\right| < 1\right\}$ and $B = \left\{z \in C : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\right\}$. Then A ∩ B is :

(A) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

(B) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

(C) an empty

(D) a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

Open solution

**Q86.** JEE Main 2022 (Online) 25th June Evening Shift

Let $z_1$ and $z_2$ be two complex numbers such that $\overline{z}_1 = i\overline{z}_2$ and $\arg\left(\frac{z_1}{z_2}\right) = \pi$. Then :

(A) $\arg z_2 = \frac{\pi}{4}$ (B) $\arg z_2 = -\frac{3\pi}{4}$

(C) $\arg z_1 = \frac{\pi}{4}$ (D) $\arg z_1 = -\frac{3\pi}{4}$

Open solution

**Q87.** JEE Main 2022 (Online) 25th June Morning Shift

Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and $z_1$ is perpendicular to the line through $z_2$ and $z_3$, then $arg(z)$ is equal to :

(A) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$ (B) $\tan^{-1}\left(\frac{24}{7}\right) - \pi$

(C) $\tan^{-1}(3) - \pi$ (D) $\tan^{-1}\left(\frac{3}{4}\right) - \pi$

Open solution

**Q88.** JEE Main 2022 (Online) 24th June Morning Shift

Let $A = \{z \in C : 1 \leq |z - (1+i)| \leq 2\}$

and $B = \{z \in A : |z - (1-i)| = 1\}$. Then, B :

(A) is an empty set (B) contains exactly two elements

(C) contains exactly three elements (D) is an infinite set

Open solution

**Q89.** JEE Main 2021 (Online) 31st August Evening Shift

If z is a complex number such that $\frac{z-i}{z-1}$ is purely imaginary, then the minimum value of | z − (3 + 3i) | is :

(A) $2\sqrt{2} - 1$ (B) $3\sqrt{2}$ (C) $6\sqrt{2}$ (D) $2\sqrt{2}$

Open solution

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**Q90.** JEE Main 2021 (Online) 27th August Morning Shift

If $S = \{z \in C : \frac{z-i}{z+2i} \in R\}$, then :

(A) S contains exactly two elements

(B) S contains only one element

(C) S is a circle in the complex plane

(D) S is a straight line in the complex plane

Open solution

**Q91.** JEE Main 2021 (Online) 26th August Evening Shift

If $\left(\sqrt{3}+i\right)^{100} = 2^{99}(p+iq)$, then p and q are roots of the equation :

(A) $x^2 - \left(\sqrt{3}-1\right)x - \sqrt{3} = 0$

(B) $x^2 + \left(\sqrt{3}+1\right)x + \sqrt{3} = 0$

(C) $x^2 + \left(\sqrt{3}-1\right)x - \sqrt{3} = 0$

(D) $x^2 - \left(\sqrt{3}+1\right)x + \sqrt{3} = 0$

Open solution

**Q92.** JEE Main 2021 (Online) 26th August Morning Shift

The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with :

(A) centre at (0, $-1$) and radius $\sqrt{2}$

(B) centre at (0, 1) and radius $\sqrt{2}$

(C) centre (0, 0) and radius $\sqrt{2}$

(D) centre at (0, 1) and radius 2

Open solution

**Q93.** JEE Main 2021 (Online) 27th July Evening Shift

Let C be the set of all complex numbers. Let

$S_1 = \{z \in C : |z - 2| \leq 1\}$ and

$S_2 = \{z \in C : z(1 + i) + \overline{z}(1 - i) \geq 4\}$.

Then, the maximum value of $\left|z - \frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to :

(A) $\frac{3+2\sqrt{2}}{4}$ (B) $\frac{5+2\sqrt{2}}{2}$ (C) $\frac{3+2\sqrt{2}}{2}$ (D) $\frac{5+2\sqrt{2}}{4}$

Open solution

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**Q94.** JEE Main 2021 (Online) 27th July Morning Shift

Let C be the set of all complex numbers. Let

$S_1 = \{z \in C : |z - 3 - 2i|^2 = 8\}$

$S_2 = \{z \in C : \text{Re}(z) \geq 5\}$ and

$S_3 = \{z \in C : |z - \overline{z}| \geq 8\}$.

Then the number of elements in $S_1 \cap S_2 \cap S_3$ is equal to :

(A) 1 (B) 0 (C) 2 (D) Infinite

Open solution

**Q95.** JEE Main 2021 (Online) 22th July Evening Shift

Let n denote the number of solutions of the equation $z^2 + 3\overline{z} = 0$, where z is a complex number. Then the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ is equal to :

(A) 1 (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) 2

Open solution

**Q96.** JEE Main 2021 (Online) 20th July Morning Shift

If z and $\omega$ are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$, then $\arg\left(\frac{1-2\overline{z}\omega}{1+3\overline{z}\omega}\right)$ is :

(Here arg(z) denotes the principal argument of complex number z)

(A) $\frac{\pi}{4}$ (B) $-\frac{3\pi}{4}$ (C) $-\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

Open solution

**Q97.** JEE Main 2021 (Online) 18th March Evening Shift

Let a complex number be w = $1 - \sqrt{3}$i. Let another complex number z be such that |zw| = 1 and arg(z) $-$ arg(w) = $\frac{\pi}{2}$. Then the area of the triangle with vertices origin, z and w is equal to :

(A) 4 (B) $\frac{1}{4}$ (C) 2 (D) $\frac{1}{2}$

Open solution

**Q98.** JEE Main 2021 (Online) 18th March Morning Shift

If the equation $a|z|^2 + \overline{\overline{\alpha}z + \alpha\overline{z}} + d = 0$ represents a circle where a, d are real constants then which of the following condition is correct?

(A) $|\alpha|^2 - ad \neq 0$ (B) $|\alpha|^2 - ad > 0$ and $a \in R - \{0\}$

(C) $|\alpha|^2 - ad \geq 0$ and $a \in R$ (D) $\alpha = 0$, $a, d \in R^+$

Open solution

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**Q99.** JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Let $S_1$, $S_2$ and $S_3$ be three sets defined as

$S_1 = \{z \in C : |z - 1| \leq \sqrt{2}\}$

$S_2 = \{z \in C : Re((1 - i)z) \geq 1\}$

$S_3 = \{z \in C : Im(z) \leq 1\}$

Then the set $S_1 \cap S_2 \cap S_3$ :

(A) has exactly three elements

(B) is a singleton

(C) has infinitely many elements

(D) has exactly two elements

**Q100.** JEE Main 2021 (Online) 17th March Morning Shift

Open solution

The area of the triangle with vertices A(z), B(iz) and C(z + iz) is :

(A) 1

(B) $\frac{1}{2}|z|^2$

(C) $\frac{1}{2}|z + iz|^2$

(D) $\frac{1}{2}$

**Q101.** JEE Main 2021 (Online) 16th March Evening Shift

Open solution

The least value of |z| where z is complex number which satisfies the inequality $\exp\left(\frac{(|z|+3)(|z|-1)}{|z|+1}\log_e 2\right) \geq \log_{\sqrt{2}}|5\sqrt{7} + 9i|, i = \sqrt{-1}$, is equal to :

(A) 8

(B) 3

(C) 2

(D) $\sqrt{5}$

**Q102.** JEE Main 2021 (Online) 16th March Morning Shift

Open solution

Let a complex number z, $|z| \neq 1$,

satisfy $\log_{\frac{1}{\sqrt{2}}}\left(\frac{|z|+11}{(|z|-1)^2}\right) \leq 2$. Then, the largest value of |z| is equal to _____.

(A) 5

(B) 8

(C) 6

(D) 7

**Q103.** JEE Main 2021 (Online) 25th February Evening Shift

Open solution

If $\alpha, \beta \in R$ are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to :

(A) $-7$

(B) 7

(C) 3

(D) $-3$

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**Q104.** JEE Main 2021 (Online) 25th February Morning Shift

Let the lines $(2 - i)z = (2 + i)\overline{z}$ and $(2 + i)z + (i - 2)\overline{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C. If the line $iz + \overline{z} + 1 + i = 0$ is tangent to this circle C, then its radius is :

(A) $\frac{3}{2\sqrt{2}}$ (B) $3\sqrt{2}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{3}{\sqrt{2}}$

Open solution

**Q105.** JEE Main 2020 (Online) 6th September Evening Slot

Let z = x + iy be a non-zero complex number such that $z^2 = i|z|^2$, where $i = \sqrt{-1}$, then z lies on the :

(A) line, y = −x (B) real axis (C) line, y = x (D) imaginary axis

Open solution

**Q106.** JEE Main 2020 (Online) 6th September Morning Slot

The region represented by
$\{z = x + iy \in C : |z| - Re(z) \leq 1\}$ is also given by the inequality :

(A) $y^2 \leq 2\left(x + \frac{1}{2}\right)$ (B) $y^2 \leq x + \frac{1}{2}$
(C) $y^2 \geq 2(x + 1)$ (D) $y^2 \geq x + 1$

Open solution

**Q107.** JEE Main 2020 (Online) 5th September Evening Slot

The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is :

(A) $-2^{15}i$ (B) $-2^{15}$ (C) $2^{15}i$ (D) $6^5$

Open solution

**Q108.** JEE Main 2020 (Online) 5th September Morning Slot

If the four complex numbers $z, \overline{z}, \overline{z} - 2\mathrm{Re}(\overline{z})$ and $z - 2Re(z)$ represent the vertices of a square of side 4 units in the Argand plane, then $|z|$ is equal to :

(A) $4\sqrt{2}$ (B) 4 (C) 2 (D) $2\sqrt{2}$

Open solution

**Q109.** JEE Main 2020 (Online) 4th September Evening Slot

If a and b are real numbers such that
$(2 + \alpha)^4 = a + b\alpha$
where $\alpha = \frac{-1+i\sqrt{3}}{2}$ then a + b is equal to :

(A) 33 (B) 9 (C) 24 (D) 57

Open solution

**Q110.** JEE Main 2020 (Online) 4th September Morning Slot

Let $u = \frac{2z+i}{z-ki}$, z = x + iy and k > 0. If the curve represented by Re(u) + Im(u) = 1 intersects the y-axis at the points P and Q where PQ = 5, then the value of k is :

(A) 2 (B) 4 (C) 1/2 (D) 3/2

Open solution

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**Q111.** JEE Main 2020 (Online) 3rd September Evening Slot

If $z_1$ , $z_2$ are complex numbers such that
$Re(z_1) = |z_1 - 1|$, $Re(z_2) = |z_2 - 1|$ , and
$\arg(z_1 - z_2) = \frac{\pi}{6}$, then $Im(z_1 + z_2)$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $2\sqrt{3}$

Open solution

**Q112.** JEE Main 2020 (Online) 2nd September Evening Slot

The imaginary part of
$\left(3+2\sqrt{-54}\right)^{\frac{1}{2}} - \left(3-2\sqrt{-54}\right)^{\frac{1}{2}}$ can be :

(A) $-2\sqrt{6}$ (B) 6 (C) $\sqrt{6}$ (D) $-\sqrt{6}$

Open solution

**Q113.** JEE Main 2020 (Online) 2nd September Morning Slot

The value of

$\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is :

(A) $\frac{1}{2}\left(\sqrt{3}-i\right)$ (B) $-\frac{1}{2}\left(\sqrt{3}-i\right)$
(C) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(1-i\sqrt{3}\right)$

Open solution

**Q114.** JEE Main 2020 (Online) 9th January Evening Slot

If z be a complex number satisfying $|Re(z)| + |Im(z)| = 4$, then $|z|$ cannot be :

(A) $\sqrt{10}$ (B) $\sqrt{7}$ (C) $\sqrt{\frac{17}{2}}$ (D) $\sqrt{8}$

Open solution

**Q115.** JEE Main 2020 (Online) 9th January Morning Slot

Let z be complex number such that
$\left|\frac{z-i}{z+2i}\right| = 1$ and $|z| = \frac{5}{2}$.
Then the value of $|z + 3i|$ is :

(A) $2\sqrt{3}$ (B) $\sqrt{10}$ (C) $\frac{15}{4}$ (D) $\frac{7}{2}$

Open solution

**Q116.** JEE Main 2020 (Online) 8th January Morning Slot

If the equation, $x^2 + bx + 45 = 0$ ($b \in R$) has conjugate complex roots and they satisfy $|z + 1| = 2\sqrt{10}$, then :

(A) $b^2 - b = 42$ (B) $b^2 + b = 12$ (C) $b^2 + b = 72$ (D) $b^2 - b = 30$

Open solution

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**Q117.** JEE Main 2020 (Online) 7th January Evening Slot

If $\frac{3+i\sin\theta}{4-i\cos\theta}$, $\theta \in [0, 2\pi]$, is a real number, then an argument of $\sin\theta + i\cos\theta$ is :

(A) $\pi - \tan^{-1}\left(\frac{3}{4}\right)$

(B) $-\tan^{-1}\left(\frac{3}{4}\right)$

(C) $\tan^{-1}\left(\frac{4}{3}\right)$

(D) $\pi - \tan^{-1}\left(\frac{4}{3}\right)$

Open solution

**Q118.** JEE Main 2020 (Online) 7th January Morning Slot

If $\mathrm{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where z = x + iy, then the point (x, y) lies on a :

(A) straight line whose slope is $\frac{3}{2}$

(B) straight line whose slope is $-\frac{2}{3}$

(C) circle whose diameter is $\frac{\sqrt{5}}{2}$

(D) circle whose centre is at $\left(-\frac{1}{2}, -\frac{3}{2}\right)$

Open solution

**Q119.** JEE Main 2019 (Online) 12th April Evening Slot

Let $z \in C$ with Im(z) = 10 and it satisfies $\frac{2z-n}{2z+n}$ = 2i - 1 for some natural number n. Then :

(A) n = 20 and Re(z) = −10

(B) n = 40 and Re(z) = 10

(C) n = 40 and Re(z) = −10

(D) n = 20 and Re(z) = 10

Open solution

**Q120.** JEE Main 2019 (Online) 12th April Morning Slot

The equation |z − i| = |z − 1|, $i = \sqrt{-1}$, represents :

(A) a circle of radius 1

(B) the line through the origin with slope − 1

(C) a circle of radius $\frac{1}{2}$

(D) the line through the origin with slope 1

Open solution

**Q121.** JEE Main 2019 (Online) 10th April Evening Slot

If z and w are two complex numbers such that |zw| = 1 and arg(z) − arg(w) = $\frac{\pi}{2}$, then :

(A) $z\overline{w} = \frac{1-i}{\sqrt{2}}$

(B) $\overline{z}w = i$

(C) $z\overline{w} = \frac{-1+i}{\sqrt{2}}$

(D) $\overline{z}w = -i$

Open solution

**Q122.** JEE Main 2019 (Online) 10th April Morning Slot

If a > 0 and z = $\frac{(1+i)^2}{a-i}$, has magnitude $\sqrt{\frac{2}{5}}$, then $\overline{z}$ is equal to :

(A) $-\frac{1}{5} + \frac{3}{5}i$

(B) $-\frac{1}{5} - \frac{3}{5}i$

(C) $\frac{1}{5} - \frac{3}{5}i$

(D) $-\frac{3}{5} - \frac{1}{5}i$

Open solution

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**Q123.** JEE Main 2019 (Online) 9th April Evening Slot

Let z ∈ C be such that |z| < 1.

If $\omega = \frac{5+3z}{5(1-z)}$, then :

(A) 4Im( $\omega$) > 5 (B) 5Im( $\omega$) < 1 (C) 5Re( $\omega$) > 4 (D) 5Re( $\omega$) > 1

Open solution

**Q124.** JEE Main 2019 (Online) 9th April Morning Slot

All the points in the set
$S = \left\{\frac{\alpha+i}{\alpha-i} : \alpha \in R\right\}(i = \sqrt{-1})$ lie on a :

(A) straight line whose slope is −1 (B) straight line whose slope is 1.

(C) circle whose radius is 1. (D) circle whose radius is $\sqrt{2}$.

Open solution

**Q125.** JEE Main 2019 (Online) 8th April Evening Slot

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}\left(i = \sqrt{-1}\right)$,

then $(1 + iz + z^5 + iz^8)^9$ is equal to :

(A) 1 (B) −1 (C) 0 (D) $(-1 + 2i)^9$

Open solution

**Q126.** JEE Main 2019 (Online) 8th April Morning Slot

If $\alpha$ and $\beta$ be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is :

(A) 2 (B) 5 (C) 4 (D) 3

Open solution

**Q127.** JEE Main 2019 (Online) 12th January Evening Slot

Let $z_1$ and $z_2$ be two complex numbers satisfying $|z_1| = 9$ and $|z_2 - 3 - 4i| = 4$. Then the minimum value of $|z_1 - z_2|$ is :

(A) 0 (B) 1 (C) 2 (D) $\sqrt{2}$

Open solution

**Q128.** JEE Main 2019 (Online) 12th January Morning Slot

If $\frac{z-\alpha}{z+\alpha}$ $(\alpha \in R)$ is a purely imaginary number and |z| = 2, then a value of $\alpha$ is :

(A) $\frac{1}{2}$ (B) $\sqrt{2}$ (C) 2 (D) 1

Open solution

**Q129.** JEE Main 2019 (Online) 11th January Evening Slot

Let z be a complex number such that |z| + z = 3 + i (where $i = \sqrt{-1}$). Then |z| is equal to :

(A) $\frac{\sqrt{34}}{3}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) $\frac{\sqrt{41}}{4}$

Open solution

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**Q130.** JEE Main 2019 (Online) 11th January Morning Slot

Let $\left(-2-\frac{1}{3}i\right)^3=\frac{x+iy}{27}\left(i=\sqrt{-1}\right)$, where x and y are real numbers, then y − x equals :

(A) − 85 (B) 85 (C) − 91 (D) 91

Open solution

**Q131.** JEE Main 2019 (Online) 10th January Evening Slot

Let $z=\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5$. If R(z) and I(z) respectively denote the real and imaginary parts of z, then :

(A) R(z) = − 3 (B) R(z) < 0 and I(z) > 0

(C) I(z) = 0 (D) R(z) > 0 and I(z) > 0

Open solution

**Q132.** JEE Main 2019 (Online) 10th January Morning Slot

Let $z_1$ and $z_2$ be any two non-zero complex numbers such that $3|z_1|=4|z_2|$. If $z=\frac{3z_1}{2z_2}+\frac{2z_2}{3z_1}$ then :

(A) $Im(z)=0$ (B) $|z|=\sqrt{\frac{17}{2}}$

(C) $|z|=\frac{1}{2}\sqrt{9+16\cos^2\theta}$ (D) Re(z) = 0

Open solution

**Q133.** JEE Main 2019 (Online) 9th January Evening Slot

Let $z_0$ be a root of the quadratic equation, $x^2 + x + 1 = 0$, If $z = 3 + 6iz_0^{81} - 3iz_0^{93}$, then arg z is equal to :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) 0

Open solution

**Q134.** JEE Main 2019 (Online) 9th January Morning Slot

Let $\alpha$ and $\beta$ be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to :

(A) -256 (B) 512 (C) -512 (D) 256

Open solution

**Q135.** JEE Main 2019 (Online) 9th January Morning Slot

Let
A = $\left\{\theta\in\left(-\frac{\pi}{2},\pi\right):\frac{3+2i\sin\theta}{1-2i\sin\theta}\,is\,purely\,imaginary\right\}$
. Then the sum of the elements in A is :

(A) $\frac{5\pi}{6}$ (B) $\pi$ (C) $\frac{3\pi}{4}$ (D) $\frac{2\pi}{3}$

Open solution

**Q136.** JEE Main 2018 (Online) 16th April Morning Slot

The least positive integer n for which $\left(\frac{1+i\sqrt{3}}{1-i\sqrt{3}}\right)^n=1$, is :

(A) 2 (B) 3 (C) 5 (D) 6

Open solution

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**Q137.** JEE Main 2018 (Offline)

If $\alpha, \beta \in C$ are the distinct roots of the equation
$x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to :

(A) 2 (B) -1 (C) 0 (D) 1

Open solution

**Q138.** JEE Main 2018 (Online) 15th April Evening Slot

If $|z - 3 + 2i| \leq 4$ then the difference between the greatest value and the least value of $|z|$ is :

(A) $2\sqrt{13}$ (B) 8 (C) $4 + \sqrt{13}$ (D) $\sqrt{13}$

Open solution

**Q139.** JEE Main 2018 (Online) 15th April Morning Slot

The set of all $\alpha \in \mathbf{R}$, for which $w = \frac{1+(1-8\alpha)z}{1-z}$ is purely imaginary number, for all $z \in \mathbf{C}$ satisfying $|z| = 1$ and $\text{Re } z \neq 1$, is :

(A) an empty set (B) {0}
(C) $\{0, \frac{1}{4}, -\frac{1}{4}\}$ (D) equal to **R**

Open solution

**Q140.** JEE Main 2017 (Online) 9th April Morning Slot

The equation
$\text{Im}\left(\frac{iz-2}{z-i}\right) + 1 = 0, z \in \mathbf{C}, z \neq i$
represents a part of a circle having radius equal to :

(A) 2 (B) 1 (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

Open solution

**Q141.** JEE Main 2017 (Online) 8th April Morning Slot

Let $z \in C$, the set of complex numbers. Then the equation, $2|z + 3i| - |z - i| = 0$ represents :

(A) a circle with radius $\frac{8}{3}$.
(B) a circle with diameter $\frac{10}{3}$.
(C) an ellipse with length of major axis $\frac{16}{3}$.
(D) an ellipse with length of minor axis $\frac{16}{9}$.

Open solution

**Q142.** JEE Main 2017 (Offline)

Let $\omega$ be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k,$$

then k is equal to :

(A) z (B) -1 (C) 1 (D) -z

Open solution

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**Q143.** JEE Main 2016 (Online) 9th April Morning Slot

The point represented by 2 + *i* in the Argand plane moves 1 unit eastwards, then 2 units northwards and finally from there $2\sqrt{2}$ units in the south-westwardsdirection. Then its new position in the Argand plane is at the point represented by :

(A) 2 + 2i (B) 1 + i (C) $-1-i$ (D) $-2-2i$

Open solution

**Q144.** JEE Main 2016 (Offline)

A value of $\theta$ for which $\frac{2+3i\sin\theta}{1-2i\ \sin\theta}$ is purely imaginary, is :

(A) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Open solution

**Q145.** JEE Main 2015 (Online) 11th April Morning Slot

If z is non-real complex number, then the minimum value of $\frac{\text{Im}\ z^5}{(\text{Im}\ z)^5}$ is :

(A) $-1$ (B) $-2$ (C) $-4$ (D) $-5$

Open solution

**Q146.** JEE Main 2015 (Online) 10th April Morning Slot

The largest value of r for which the region represented by the set

$\{\omega \in C/|\omega-4-i|\leq r\}$

is contained in the region represented by the set

$\{z \in C/|z-1|\leq|z+i|\}$, is equal to :

(A) $\sqrt{17}$ (B) $2\sqrt{2}$ (C) $\frac{3}{2}\sqrt{2}$ (D) $\frac{5}{2}\sqrt{2}$

Open solution

**Q147.** JEE Main 2015 (Offline)

A complex number z is said to be unimodular if $|z|=1$. Suppose $z_1$ and $z_2$ are complex numbers such that $\frac{z_1-2z_2}{2-z_1\bar{z}_2}$ is unimodular and $z_2$ is not unimodular. Then the point $z_1$ lies on a :

(A) circle of radius 2. (B) circle of radius $\sqrt{2}$.

(C) straight line parallel to x-axis (D) straight line parallel to y-axis.

Open solution

**Q148.** JEE Main 2014 (Online) 19th April Morning Slot

For all complex numbers z of the form 1 + i$\alpha$, $\alpha \in$ **R**, if $z^2$ = x + iy, then :

(A) $y^2-4x+2=0$ (B) $y^2+4x-4=0$ (C) $y^2-4x+4=0$ (D) $y^2+4x+2=0$

Open solution

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**Q149.** JEE Main 2014 (Online) 12th April Morning Slot

Let z $\neq -$ i be any complex number such that $\frac{z-i}{z+i}$ is a purely imaginary number.

Then z + $\frac{1}{z}$ is :

(A) 0

(B) any non-zero real number other than 1.

(C) any non-zero real number.

(D) a purely imaginary number.

Open solution

**Q150.** JEE Main 2014 (Online) 11th April Morning Slot

If $z_1$, $z_2$ and $z_3$, $z_4$ are 2 pairs of complex conjugate numbers, then $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ equals :

(A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) $\pi$

Open solution

**Q151.** JEE Main 2014 (Online) 9th April Morning Slot

Let w (Im w $\neq$ 0) be a complex number. Then the set of all complex numbers z satisfying the equation w $-\ \overline{w}$ z = k (1 $-$ z), for some real number k, is :

(A) $\{z : |z| = 1\}$

(B) $\{z : z = \overline{z}\}$

(C) $\{z : z \neq 1\}$

(D) $\{z : |z| = 1, z \neq 1\}$

Open solution

**Q152.** JEE Main 2014 (Offline)

If z is a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$ :

(A) is strictly greater that $\frac{5}{2}$

(B) is strictly greater that $\frac{3}{2}$ but less than $\frac{5}{2}$

(C) is equal to $\frac{5}{2}$

(D) lie in the interval (1, 2)

Open solution

**Q153.** JEE Main 2013 (Online) 25th April Morning Slot

Let z satisfy $|z| = 1$ and z = 1 - $\bar{z}$

**Statement 1:** z is a real number.

**Statement 2:** Principal argument of z is $\frac{\pi}{3}$.

(A) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1.

(B) Statement 1 is false; Statement 2 is true;

(C) Statement 1 is true; Statement 2 is false;

(D) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

Open solution

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**Q154.** JEE Main 2013 (Online) 23th April Morning Slot

Let a = Im $\left(\frac{1+z^2}{2iz}\right)$, where z is any non-zero complex number.
The set A = {a: |z| = 1 and z $\neq \pm$ 1} is equal to:

(A) (-1, 1) (B) [-1, 1] (C) [0, 1) (D) (-1, 0]

Open solution

**Q155.** JEE Main 2013 (Online) 22th April Morning Slot

If a complex number z satisfies the equation z + $\sqrt{2}|z+1|$ + i = 0, then $|z|$ is equal to:

(A) 2 (B) $\sqrt{3}$ (C) $\sqrt{5}$ (D) 1

Open solution

**Q156.** JEE Main 2013 (Online) 9th April Morning Slot

If $Z_1 \neq 0$ and $Z_2$ be two complex numbers such that $\frac{Z_2}{Z_1}$ is a purely imaginary number, then $\left|\frac{2Z_1+3Z_2}{2Z_1-3Z_2}\right|$ is equal to:

(A) 2 (B) 5 (C) 3 (D) 1

Open solution

**Q157.** JEE Main 2013 (Offline)

If z is a complex number of unit modulus and argument $\theta$, then arg $\left(\frac{1+z}{1+\bar{z}}\right)$ equals :

(A) $-\theta$ (B) $\frac{\pi}{2}-\theta$ (C) $\theta$ (D) $\pi-\theta$

Open solution

**Q158.** AIEEE 2012 (Online) 26th May

$|z_1+z_2|^2+|z_1-z_2|^2$ is equal to

(A) $2\left(|z_1|+|z_2|\right)$
(B) $2\left(|z_1|^2+|z_2|^2\right)$
(C) $|z_1|\,|z_2|$
(D) $|z_1|^2+|z_2|^2$

Open solution

**Q159.** AIEEE 2012 (Online) 19th May

Let $Z$ and $W$ be complex numbers such that $|Z|=|W|$, and $\arg Z$ denotes the principal argument of $Z$.
Statement 1: If $\arg Z+\arg W=\pi$, then $Z=-\bar{W}$.
Statement 2: $|Z|=|W|$, implies $\arg Z-\arg \bar{W}=\pi$.

(A) Statement 1 is true, Statement 2 is false.
(B) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1.
(C) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.
(D) Statement 1 is false, Statement 2 is true.

Open solution

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**Q160.** AIEEE 2012 (Online) 12th May

The area of the triangle whose vertices are complex numbers $z, iz, z+iz$ in the Argand diagram is

(A) $2|z|^2$ (B) $\frac{1}{2}|z|^2$ (C) $4|z|^2$ (D) $|z|^2$

Open solution

**Q161.** AIEEE 2012 (Online) 7th May

Let $Z_1$ and $Z_2$ be any two complex number.
Statement 1: $|Z_1 - Z_2| \geq |Z_1| - |Z_2|$
Statement 2: $|Z_1 + Z_2| \leq |Z_1| + |Z_2|$

(A) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is false, Statement 2 is true.

Open solution

**Q162.** AIEEE 2012

If $z \neq 1$ and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies :

(A) either on the real axis or a circle passing through the origin.

(B) on a circle with centre at the origin

(C) either on real axis or on a circle not passing through the origin.

(D) on the imaginary axis.

Open solution

**Q163.** AIEEE 2011

Let $\alpha$ , $\beta$ be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line Re z = 1, then it is necessary that :

(A) $\beta \in (-1, 0)$ (B) $|\beta| = 1$ (C) $\beta \in (1, \infty)$ (D) $\beta \in (0, 1)$

Open solution

**Q164.** AIEEE 2011

If $\omega(\neq 1)$ is a cube root of unity, and $(1+\omega)^7 = A + B\omega$ . Then $(A, B)$ equals :

(A) (1 ,1) (B) (1, 0) (C) (- 1 ,1) (D) (0 ,1)

Open solution

**Q165.** AIEEE 2010

The number of complex numbers z such that $|z-1| = |z+1| = |z-i|$ equals :

(A) 1 (B) 2 (C) $\infty$ (D) 0

Open solution

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**Q166.** AIEEE 2009

If $\left|z-\frac{4}{z}\right|=2$, then the maximum value of $|z|$ is equal to :

(A) $\sqrt{5}+1$ (B) 2 (C) $2+\sqrt{2}$ (D) $\sqrt{3}+1$

Open solution

**Q167.** AIEEE 2008

The conjugate of a complex number is $\frac{1}{i-1}$ then that complex number is :

(A) $\frac{-1}{i-1}$ (B) $\frac{1}{i+1}$ (C) $\frac{-1}{i+1}$ (D) $\frac{1}{i-1}$

Open solution

**Q168.** AIEEE 2007

If $|z+4| \leq 3$, then the maximum value of $|z+1|$ is :

(A) 6 (B) 0 (C) 4 (D) 10

Open solution

**Q169.** AIEEE 2006

The value of $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11}+i\cos\frac{2k\pi}{11}\right)$ is :

(A) i (B) 1 (C) - 1 (D) - i

Open solution

**Q170.** AIEEE 2006

If $z^2+z+1=0$, where z is a complex number, then the value of $\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+\ldots\ldots\ldots+\left(z^6+\frac{1}{z^6}\right)^2$ is :

(A) 18 (B) 54 (C) 6 (D) 12

Open solution

**Q171.** AIEEE 2005

If the cube roots of unity are 1, $\omega$, $\omega^2$ then the roots of the equation $(x-1)^3$ + 8 = 0, are :

(A) $-1,-1+2\,\omega,-1-2\,\omega^2$ (B) $-1,-1,-1$

(C) $-1,1-2\omega,1-2\omega^2$ (D) $-1,1+2\omega,1+2\omega^2$

Open solution

**Q172.** AIEEE 2005

If $z_1$ and $z_2$ are two non-zero complex numbers such that $|z_1+z_2|=|z_1|+|z_2|$, then arg $z_1$ - arg $z_2$ is equal to :

(A) $\frac{\pi}{2}$ (B) $-\pi$ (C) 0 (D) $\frac{-\pi}{2}$

Open solution

**Q173.** AIEEE 2005

If $\omega=\frac{z}{z-\frac{1}{3}i}$ and $|\omega|=1$, then $z$ lies on :

(A) an ellipse (B) a circle (C) a straight line (D) a parabola

Open solution

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**Q174.** AIEEE 2004

Let z and w be complex numbers such that $\overline{z} + i\overline{w} = 0$ and arg zw = $\pi$. Then arg z equals :

(A) $\frac{5\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$

Open solution

**Q175.** AIEEE 2004

If $z = x - iy$ and $z^{\frac{1}{3}} = p + iq$, then

$\frac{\left(\frac{x}{p} + \frac{y}{q}\right)}{(p^2+q^2)}$ is equal to :

(A) - 2 (B) - 1 (C) 2 (D) 1

Open solution

**Q176.** AIEEE 2004

If $|z^2 - 1| = |z|^2 + 1$, then z lies on :

(A) an ellipse (B) the imaginary axis

(C) a circle (D) the real axis

Open solution

**Q177.** AIEEE 2003

If $z$ and $\omega$ are two non-zero complex numbers such that $|z\omega| = 1$ and $Arg(z) - Arg(\omega) = \frac{\pi}{2}$, then $\overline{z}\omega$ is equal to

(A) $-i$ (B) 1 (C) - 1 (D) $i$

Open solution

**Q178.** AIEEE 2003

Let $Z_1$ and $Z_2$ be two roots of the equation $Z^2 + aZ + b = 0$, Z being complex. Further , assume that the origin, $Z_1$ and $Z_2$ form an equilateral triangle. Then :

(A) $a^2 = 4b$ (B) $a^2 = b$ (C) $a^2 = 2b$ (D) $a^2 = 3b$

Open solution

**Q179.** AIEEE 2003

If $\left(\frac{1+i}{1-i}\right)^x = 1$ then :

(A) x = 2n + 1, where n is any positive integer

(B) x = 4n , where n is any positive integer

(C) x = 2n, where n is any positive integer

(D) x = 4n + 1, where n is any positive integer.

Open solution

**Q180.** AIEEE 2002

z and w are two nonzero complex numbers such that $|z| = |w|$ and Arg z + Arg w =$\pi$ then z equals

(A) $\overline{\omega}$ (B) $-\overline{\omega}$ (C) $\omega$ (D) $-\omega$

Open solution

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**Q181.** AIEEE 2002

If $|z-4| < |z-2|$, its solution is given by :

(A) $\mathrm{Re}(z) > 0$ (B) $\mathrm{Re}(z) < 0$

(C) $\mathrm{Re}(z) > 3$ (D) $\mathrm{Re}(z) > 2$

Open solution

**Q182.** AIEEE 2002

The locus of the centre of a circle which touches the circle $|z-z_1| = a$ and $|z-z_2| = b$ externally

($z$, $z_1$ & $z_2$ are complex numbers) will be :

(A) an ellipse (B) a hyperbola (C) a circle (D) none of these

Open solution

## Numerical 38 questions

**Q183.** JEE Main 2026 (Online) 24th January Evening Shift

Let $z = (1+i)(1+2i)(1+3i)\ldots(1+ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then $n$ is equal to ____

Open solution

**Q184.** JEE Main 2026 (Online) 22nd January Morning Shift

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $\beta = \frac{-1-i\sqrt{3}}{2}, i = \sqrt{-1}$. If

$(7-7\alpha+9\beta)^{20} + (9+7\alpha-7\beta)^{20} + (-7+9\alpha+7\beta)^{20} + (14+7\alpha+7\beta)^{20} = m^{10}$,

then $m$ is ____

Open solution

**Q185.** JEE Main 2025 (Online) 4th April Evening Shift

If $\alpha$ is a root of the equation $x^2+x+1=0$ and $\sum_{k=1}^{n}\left(\alpha^k + \frac{1}{\alpha^k}\right)^2 = 20$, then n is equal to _____.

Open solution

**Q186.** JEE Main 2025 (Online) 4th April Morning Shift

Let $A = \{z \in C : |z-2-i| = 3\}, B = \{z \in C : \mathrm{Re}(z-iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z\in S}|z|^2$ is equal to _____.

Open solution

**Q187.** JEE Main 2025 (Online) 29th January Evening Shift

Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right| = 1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in C$, where $\omega$ and $\omega^2$ are the roots of $x^2+x+1=0$, is equal to _____ .

Open solution

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**Q188.** JEE Main 2025 (Online) 23rd January Evening Shift

Let $\alpha, \beta$ be the roots of the equation $x^2 - ax - b = 0$ with $\mathrm{Im}(\alpha) < \mathrm{Im}(\beta)$. Let $P_n = \alpha^n - \beta^n$. If $P_3 = -5\sqrt{7}i, P_4 = -3\sqrt{7}i, P_5 = 11\sqrt{7}i$ and $P_6 = 45\sqrt{7}i$, then $|\alpha^4 + \beta^4|$ is equal to _____.

Open solution

**Q189.** JEE Main 2024 (Online) 9th April Morning Shift

The sum of the square of the modulus of the elements in the set $\{z = a + ib : a, b \in \mathbf{Z}, z \in \mathbf{C}, |z-1| \leq 1, |z-5| \leq |z-5i|\}$ is _____.

Open solution

**Q190.** JEE Main 2024 (Online) 1st February Morning Shift

Let $P = \{z \in \mathbb{C} : |z+2-3i| \leq 1\}$ and $Q = \{z \in \mathbb{C} : z(1+i) + \bar{z}(1-i) \leq -8\}$. Let in $P \cap Q$, $|z-3+2i|$ be maximum and minimum at $z_1$ and $z_2$ respectively. If $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$, where $\alpha, \beta$ are integers, then $\alpha + \beta$ equals _____.

Open solution

**Q191.** JEE Main 2024 (Online) 31st January Morning Shift

If $\alpha$ denotes the number of solutions of $|1-i|^x = 2^x$ and $\beta = \left(\frac{|z|}{\arg(z)}\right)$, where $z = \frac{\pi}{4}(1+i)^4 \left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right], i = \sqrt{-1}$, then the distance of the point $(\alpha, \beta)$ from the line $4x - 3y = 7$ is _____.

Open solution

**Q192.** JEE Main 2024 (Online) 29th January Evening Shift

Let $\alpha, \beta$ be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\mathrm{Im}(\alpha) > \mathrm{Im}(\beta)$. Let $a, b$ be integers not divisible by 3 and $n$ be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a+ib), i = \sqrt{-1}$. Then $n + a + b$ is equal to _____.

Open solution

**Q193.** JEE Main 2024 (Online) 29th January Morning Shift

Let $\alpha, \beta$ be the roots of the equation $x^2 - x + 2 = 0$ with $\mathrm{Im}(\alpha) > \mathrm{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____.

Open solution

**Q194.** JEE Main 2024 (Online) 27th January Evening Shift

Let the complex numbers $\alpha$ and $\frac{1}{\bar{\alpha}}$ lie on the circles $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100|\alpha|^2$ is _____.

Open solution

**Q195.** JEE Main 2024 (Online) 27th January Morning Shift

If $\alpha$ satisfies the equation $x^2 + x + 1 = 0$ and $(1+\alpha)^7 = A + B\alpha + C\alpha^2, A, B, C \geqslant 0$, then $5(3A - 2B - C)$ is equal to _____.

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**Q196.** JEE Main 2023 (Online) 13th April Morning Shift

Let $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$. Let $\mathrm{Re}(w) = 0$ be the circle C of radius 1 in the first quadrant touching the line $y = 1$ and the $y$-axis. If the curve $\mathrm{Im}(w) = 0$ intersects C at A and B, then $30(AB)^2$ is equal to _____

Open solution

**Q197.** JEE Main 2023 (Online) 11th April Evening Shift

Let $S = \left\{z \in \mathbb{C} - \{i, 2i\} : \frac{z^2+8iz-15}{z^2-3iz-2} \in \mathbb{R}\right\}$. If $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$, then $242\alpha^2$ is equal to _____.

Open solution

**Q198.** JEE Main 2023 (Online) 6th April Evening Shift

For $\alpha, \beta, z \in \mathbb{C}$ and $\lambda > 1$, if $\sqrt{\lambda - 1}$ is the radius of the circle $|z-\alpha|^2 + |z-\beta|^2 = 2\lambda$, then $|\alpha - \beta|$ is equal to _____.

Open solution

**Q199.** JEE Main 2023 (Online) 30th January Morning Shift

Let $z = 1 + i$ and $z_1 = \frac{1+i\bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$. Then $\frac{12}{\pi}\arg(z_1)$ is equal to _____.

Open solution

**Q200.** JEE Main 2023 (Online) 29th January Evening Shift

Let $\alpha = 8 - 14i, A = \left\{z \in \mathbb{C} : \frac{\alpha z - \overline{\alpha z}}{z^2 - (\bar{z})^2 - 112i} = 1\right\}$ and $B = \{z \in \mathbb{C} : |z + 3i| = 4\}$. Then $\sum_{z \in A \cap B} (\mathrm{Re}z - \mathrm{Im}z)$ is equal to _____.

Open solution

**Q201.** JEE Main 2022 (Online) 28th July Evening Shift

Let $z = a + ib, b \neq 0$ be complex numbers satisfying $z^2 = \bar{z} \cdot 2^{1-|z|}$. Then the least value of $n \in N$, such that $z^n = (z+1)^n$, is equal to _____.

Open solution

**Q202.** JEE Main 2022 (Online) 27th July Morning Shift

Let $S = \{z \in \mathbb{C} : z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} (\mathrm{Re}(z) + \mathrm{Im}(z))$ is equal to _____.

Open solution

**Q203.** JEE Main 2022 (Online) 29th June Morning Shift

Let $S = \{z \in C : |z-2| \leq 1, z(1+i) + \bar{z}(1-i) \leq 2\}$. Let $|z - 4i|$ attain minimum and maximum values, respectively, at $z_1 \in S$ and $z_2 \in S$. If $5(|z_1|^2 + |z_2|^2) = \alpha + \beta\sqrt{5}$, where $\alpha$ and $\beta$ are integers, then the value of $\alpha + \beta$ is equal to _____.

Open solution

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**Q204.** JEE Main 2022 (Online) 28th June Evening Shift

Sum of squares of modulus of all the complex numbers z satisfying $\overline{z} = iz^2 + z^2 - z$ is equal to _____.

Open solution

**Q205.** JEE Main 2022 (Online) 28th June Morning Shift

The number of elements in the set {z = a + ib $\in$ C : a, b $\in$ Z and 1 < | z $-$ 3 + 2i | < 4} is _____.

Open solution

**Q206.** JEE Main 2022 (Online) 26th June Evening Shift

If $z^2 + z + 1 = 0, z \in C$, then

$\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$ is equal to _____.

Open solution

**Q207.** JEE Main 2022 (Online) 24th June Evening Shift

Let S = {z $\in$ C : |z $-$ 3| $\leq$ 1 and z(4 + 3i) + $\overline{z}$(4 $-$ 3i) $\leq$ 24}. If $\alpha$ + i$\beta$ is the point in S which is closest to 4i, then 25($\alpha$ + $\beta$) is equal to _____.

Open solution

**Q208.** JEE Main 2021 (Online) 1st September Evening Shift

If for the complex numbers z satisfying | z $-$ 2 $-$ 2i | $\leq$ 1, the maximum value of | 3iz + 6 | is attained at a + ib, then a + b is equal to _____.

Open solution

**Q209.** JEE Main 2021 (Online) 31st August Morning Shift

A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$, then the minimum value of $\left|z - 9\sqrt{2} - 2i\right|^2$ is equal to _____.

Open solution

**Q210.** JEE Main 2021 (Online) 27th August Evening Shift

Let $z_1$ and $z_2$ be two complex numbers such that $\arg(z_1 - z_2) = \frac{\pi}{4}$ and $z_1$, $z_2$ satisfy the equation | z $-$ 3 | = Re(z). Then the imaginary part of $z_1 + z_2$ is equal to _____.

Open solution

**Q211.** JEE Main 2021 (Online) 26th August Evening Shift

The least positive integer n such that $\frac{(2i)^n}{(1-i)^{n-2}}, i = \sqrt{-1}$ is a positive integer, is _____.

Open solution

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**Q212.** JEE Main 2021 (Online) 26th August Morning Shift

Let $z = \frac{1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. Then the value of $21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \ldots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$ is _____.

Open solution

**Q213.** JEE Main 2021 (Online) 27th July Evening Shift

If the real part of the complex number $z = \frac{3+2i\cos\theta}{1-3i\cos\theta}, \theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos^2\theta$ is equal to _____.

Open solution

**Q214.** JEE Main 2021 (Online) 25th July Evening Shift

The equation of a circle is $\mathrm{Re}(z^2) + 2(\mathrm{Im}(z))^2 + 2\mathrm{Re}(z) = 0$, where $z = x + iy$. A line which passes through the center of the given circle and the vertex of the parabola, $x^2 - 6x - y + 13 = 0$, has y-intercept equal to _____.

Open solution

**Q215.** JEE Main 2021 (Online) 18th March Morning Shift

Let $z_1, z_2$ be the roots of the equation $z^2 + az + 12 = 0$ and $z_1, z_2$ form an equilateral triangle with origin. Then, the value of $|a|$ is :

Open solution

**Q216.** JEE Main 2021 (Online) 16th March Morning Shift

Let z and $\omega$ be two complex numbers such that $\omega = z\bar{z} - 2z + 2$, $\left|\frac{z+i}{z-3i}\right| = 1$ and $\mathrm{Re}(\omega)$ has minimum value. Then, the minimum value of $n \in N$ for which $\omega^n$ is real, is equal to _____.

Open solution

**Q217.** JEE Main 2021 (Online) 26th February Evening Shift

Let z be those complex numbers which satisfy

$|z + 5| \leq 4$ and $z(1 + i) + \bar{z}(1 - i) \geq -10$, $i = \sqrt{-1}$.

If the maximum value of $|z + 1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is _____.

Open solution

**Q218.** JEE Main 2021 (Online) 24th February Evening Shift

Let $i = \sqrt{-1}$. If $\frac{\left(-1+i\sqrt{3}\right)^{21}}{(1-i)^{24}} + \frac{\left(1+i\sqrt{3}\right)^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to _____.

Open solution

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**Q219.** JEE Main 2021 (Online) 24th February Morning Shift

If the least and the largest real values of $\alpha$, for which the equation $z + \alpha|z - 1| + 2i = 0$ ($z \in C$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to _____.

Open solution

**Q220.** JEE Main 2020 (Online) 3rd September Morning Slot

If $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{1+i}{i-1}\right)^{\frac{n}{3}} = 1$, (m, n $\in$ N) then the greatest common divisor of the least values of m and n is _____ .

Open solution

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# Answer Key

## MCQ (Single Correct Answer)

| | | | | |
|---|---|---|---|---|
| 1. **B** | 2. **A** | 3. **A** | 4. **D** | 5. **B** |
| 6. **D** | 7. **A** | 8. **C** | 9. **D** | 10. **A** |
| 11. **D** | 12. **C** | 13. **D** | 14. **D** | 15. **D** |
| 16. **A** | 17. **B** | 18. **A** | 19. **D** | 20. **C** |
| 21. **A** | 22. **A** | 23. **B** | 24. **A** | 25. **C** |
| 26. **A** | 27. **D** | 28. **A** | 29. **A** | 30. **B** |
| 31. **C** | 32. **B** | 33. **B** | 34. **B** | 35. **A** |
| 36. **A** | 37. **C** | 38. **C** | 39. **B** | 40. **B** |
| 41. **A** | 42. **C** | 43. **D** | 44. **D** | 45. **B** |
| 46. **D** | 47. **B** | 48. **C** | 49. **A** | 50. **D** |
| 51. **B** | 52. **B** | 53. **A** | 54. **C** | 55. **B** |
| 56. **C** | 57. **D** | 58. **D** | 59. **D** | 60. **C** |
| 61. **A** | 62. **D** | 63. **A** | 64. **D** | 65. **A** |
| 66. **D** | 67. **B** | 68. **B** | 69. **D** | 70. **D** |
| 71. **B** | 72. **A** | 73. **C** | 74. **C** | 75. **A** |
| 76. **C** | 77. **D** | 78. **C** | 79. **D** | 80. **D** |
| 81. **C** | 82. **A** | 83. **C** | 84. **A** | 85. **B** |
| 86. **C** | 87. **B** | 88. **D** | 89. **D** | 90. **D** |
| 91. **A** | 92. **B** | 93. **D** | 94. **A** | 95. **B** |
| 96. **B** | 97. **D** | 98. **B** | 99. **C** | 100. **B** |
| 101. **B** | 102. **D** | 103. **A** | 104. **A** | 105. **C** |
| 106. **A** | 107. **A** | 108. **D** | 109. **B** | 110. **A** |
| 111. **D** | 112. **A** | 113. **B** | 114. **B** | 115. **D** |
| 116. **D** | 117. **D** | 118. **C** | 119. **C** | 120. **D** |
| 121. **D** | 122. **B** | 123. **D** | 124. **C** | 125. **B** |
| 126. **C** | 127. **A** | 128. **C** | 129. **B** | 130. **D** |
| 131. **C** | 132. **C** | 133. **A** | 134. **A** | 135. **D** |
| 136. **B** | 137. **D** | 138. **A** | 139. **B** | 140. **C** |
| 141. **A** | 142. **D** | 143. **B** | 144. **B** | 145. **C** |
| 146. **D** | 147. **A** | 148. **B** | 149. **C** | 150. **A** |

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| | | | | |
|---|---|---|---|---|
| 151. **D** | 152. **B** | 153. **B** | 154. **A** | 155. **C** |
| 156. **D** | 157. **C** | 158. **B** | 159. **A** | 160. **B** |
| 161. **B** | 162. **A** | 163. **C** | 164. **A** | 165. **A** |
| 166. **A** | 167. **C** | 168. **A** | 169. **D** | 170. **D** |
| 171. **C** | 172. **C** | 173. **C** | 174. **C** | 175. **A** |
| 176. **B** | 177. **A** | 178. **D** | 179. **B** | 180. **B** |
| 181. **C** | 182. **B** | | | |

## Numerical

| | | | | |
|---|---|---|---|---|
| 183. **5** | 184. **49** | 185. **11** | 186. **22** | 187. **10** |
| 188. **31** | 189. **9** | 190. **36** | 191. **3** | 192. **49** |
| 193. **13** | 194. **20** | 195. **5** | 196. **24** | 197. **1680** |
| 198. **2** | 199. **9** | 200. **14** | 201. **6** | 202. **0** |
| 203. **26** | 204. **2** | 205. **40** | 206. **2** | 207. **80** |
| 208. **5** | 209. **98** | 210. **6** | 211. **6** | 212. **13** |
| 213. **1** | 214. **1** | 215. **6** | 216. **4** | 217. **48** |
| 218. **310** | 219. **10** | 220. **4** | | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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